

EDA - POWER BI

WEEK - 1

REPORT

SUBMITTED BY

Divya Dharshni P

E25DA014

Week 1:

Superstore Sales Data Import, Power Query & Data Cleaning

Dataset:

Superstore Sales Dataset

Problem Statement:

A retail organization wants to analyze its sales data to understand customer purchases, product categories, regional performance, and profitability. Before creating a business intelligence dashboard, the raw Superstore dataset must be imported, cleaned, transformed, and prepared in Power BI.

Objectives:

Import data into Power BI Desktop and navigate Report, Data, and Model views.

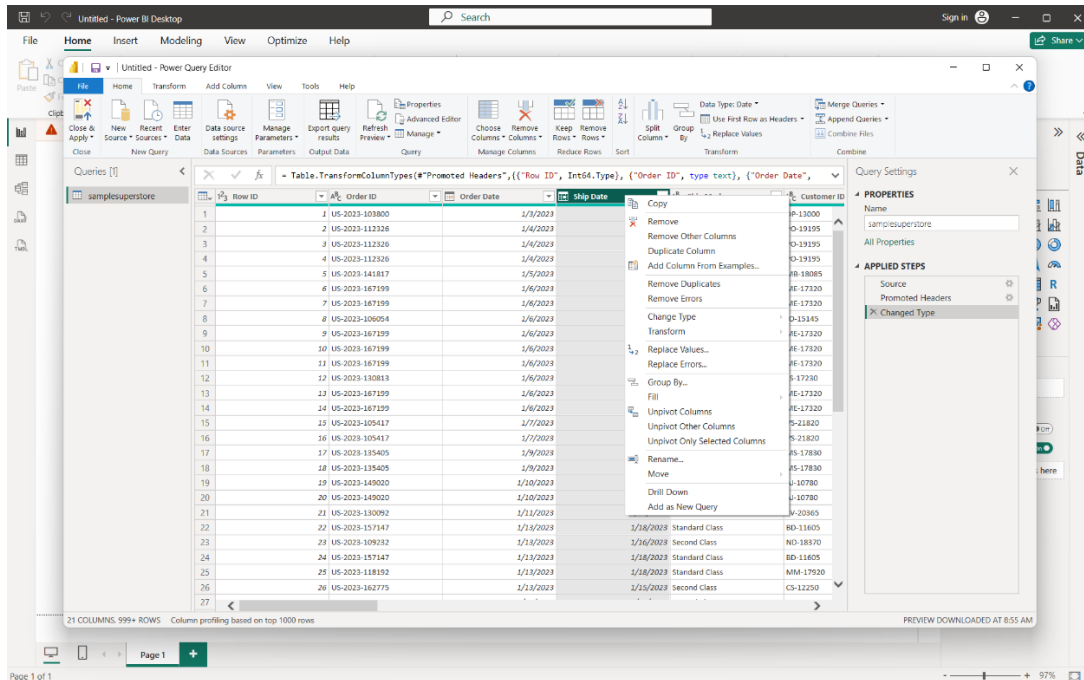
Inspect dataset attributes using Power Query Editor; verify data types, missing values, and record uniqueness.

Perform column transformations (rename, split, text trimming, data type coercion).

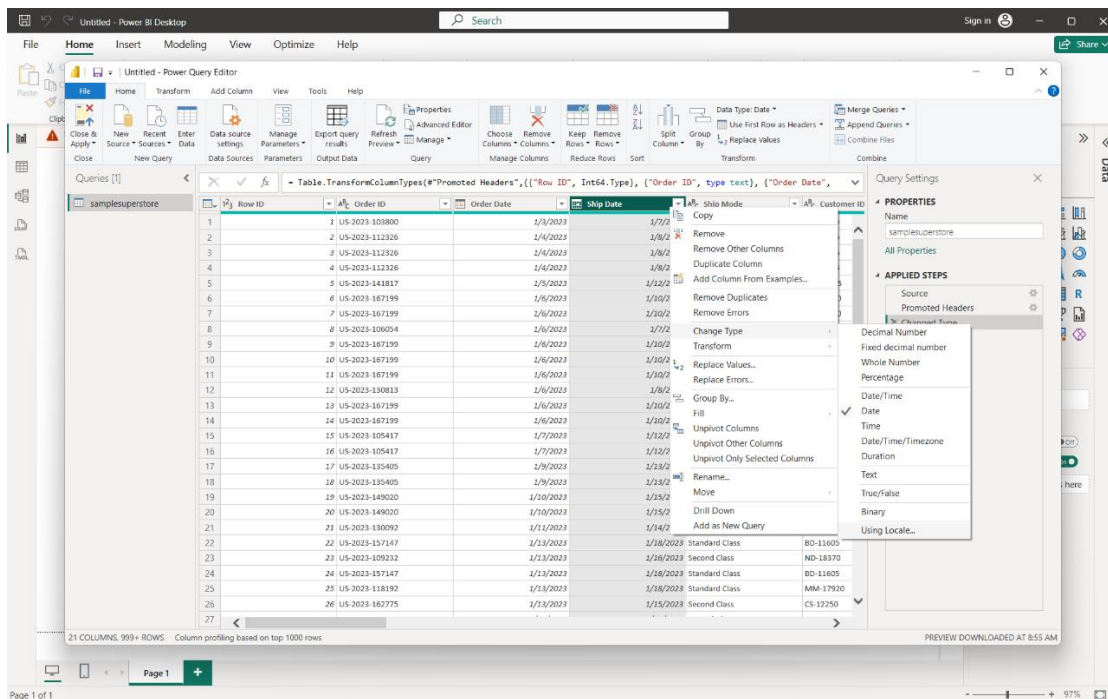
Extract temporal attributes from Order Date (Year, Quarter, Month, Day, Day of Week).

Tasks & Deliverables:

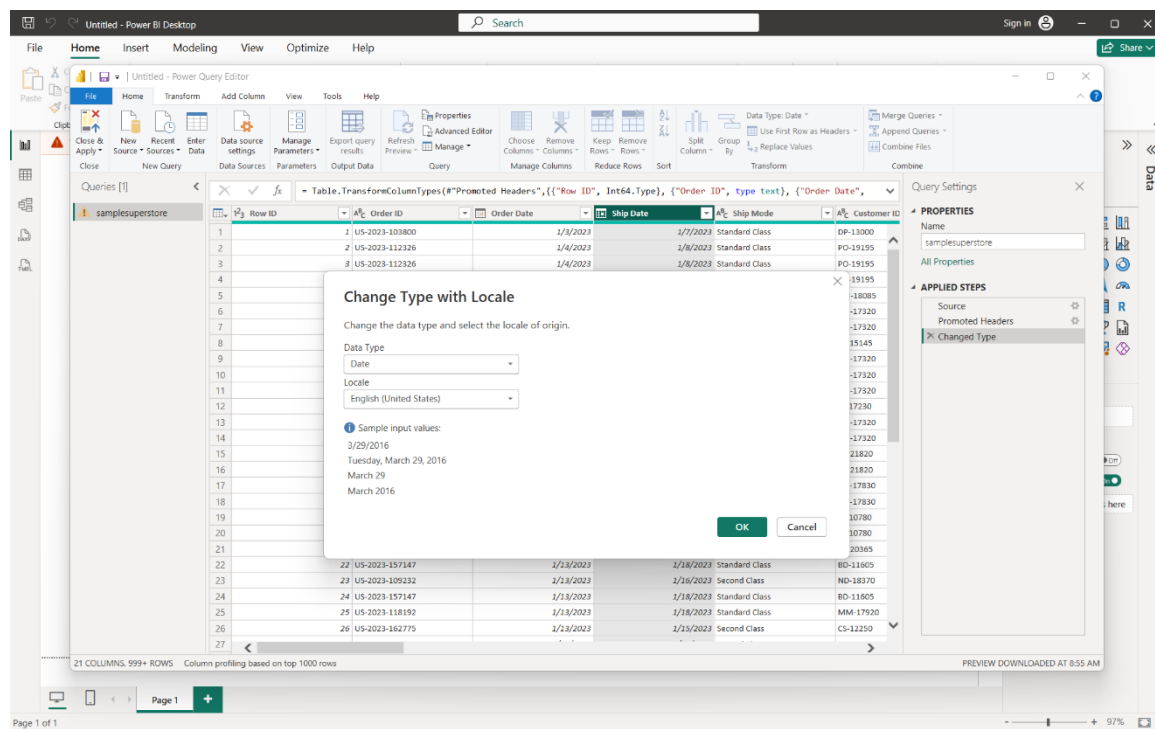
1. Load Superstore CSV into Power Query Editor.



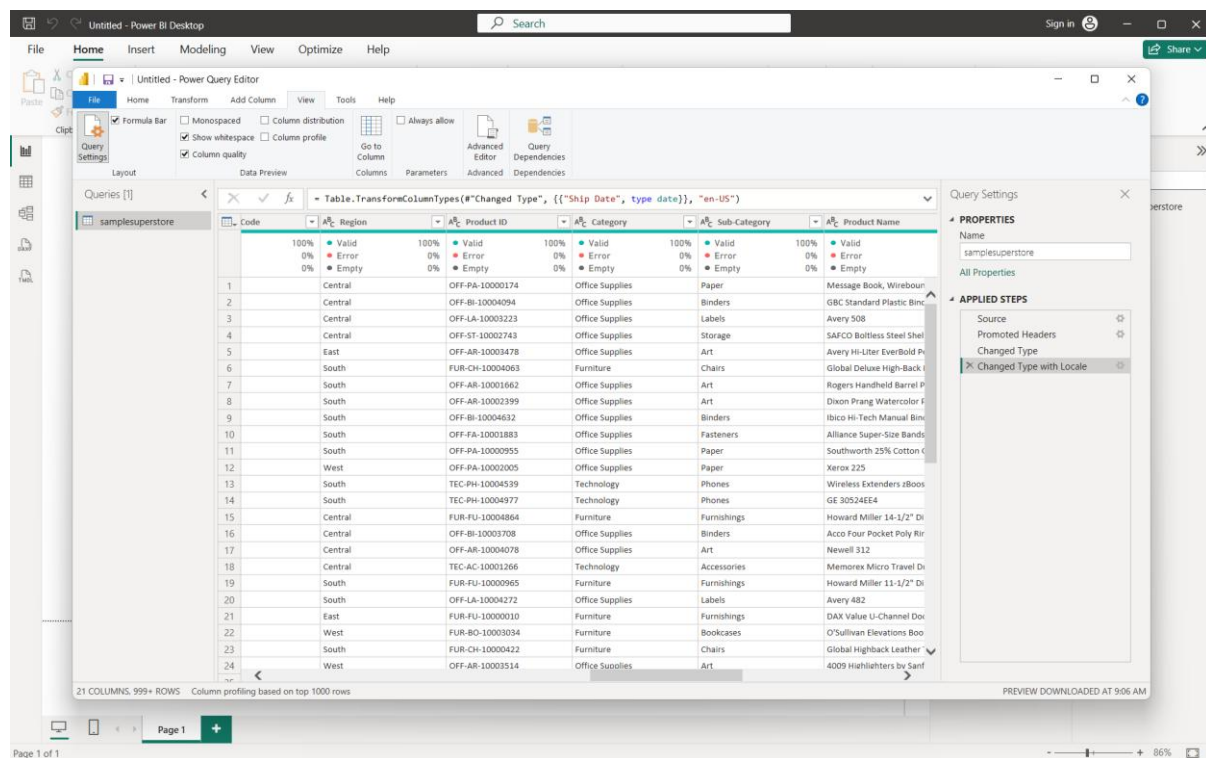
2. Standardize text formatting across categorical attributes (Category, Sub-Category, Segment).



3. Perform column transformations (rename, split, text trimming, data type coercion).



4. Extract temporal attributes from Order Date (Year, Quarter, Month, Day, Day of Week)."



5. Add custom date columns using Power Query M code / UI transformations.

The screenshot shows the Power Query Editor interface. The main area displays a table with 21 columns and 999+ rows. The columns are: Row ID, Order ID, Order Date, Ship Date, Ship Mode, and Customer ID. The 'Order Date' and 'Ship Date' columns are highlighted in blue, indicating they are date columns. The 'Query Settings' pane on the right shows the 'Applied Steps' list, which includes 'Source', 'Promoted Headers', 'Changed Type', 'Changed Type with Locale', and 'Trimmed Text'.

6. & Apply transformations to load a clean data model

The screenshot shows the Power Query Editor interface. The main area displays a table with 25 columns and 999+ rows. The columns are: Profit, Year, Day of Year, Month Name, and Quarter. The 'Month Name' and 'Quarter' columns are highlighted in blue, indicating they are date columns. The 'Query Settings' pane on the right shows the 'Applied Steps' list, which includes 'Source', 'Promoted Headers', 'Changed Type', 'Changed Type with Locale', 'Trimmed Text', 'Trimmed Text1', 'Trimmed Text2', 'Inserted Year', 'Inserted Day of Year', 'Inserted Month Name', and 'Inserted Quarter'.

Conclusion:

The Superstore Sales dataset was successfully imported and cleaned using Power Query in Power BI. The data types were verified, unnecessary inconsistencies were handled, and required column transformations were performed. Date-related attributes were also extracted from the Order Date column. The cleaned dataset is now ready for further analysis and visualization in Power BI.